

L4 - Transformations of $\frac{1}{x}$ - Lesson

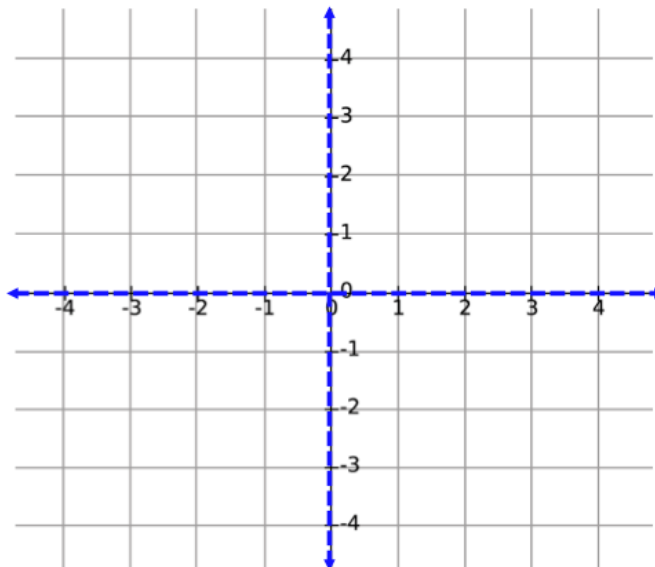
MCR3U

Base Function:

Graph of Base Function:

Key Points:

x	y



Asymptotes

Asymptote:

The function $f(x) = \frac{1}{x}$ has two asymptotes:

Vertical Asymptote: Division by zero is undefined. Therefore the expression in the denominator of the function can not be zero. Therefore $x \neq 0$. This is why the vertical line $x = 0$ is an asymptote for this function.

Horizontal Asymptote: For the range, there can never be a situation where the result of the division is zero. Therefore the line $y = 0$ is a horizontal asymptote. For all functions where the denominator is a higher degree than the numerator, there will be a horizontal asymptote at $y = 0$.

Example 1: Describe the combination of transformations that must be applied to the base function $f(x) = \frac{1}{x}$ to obtain the transformed function. Then, write the corresponding equations.

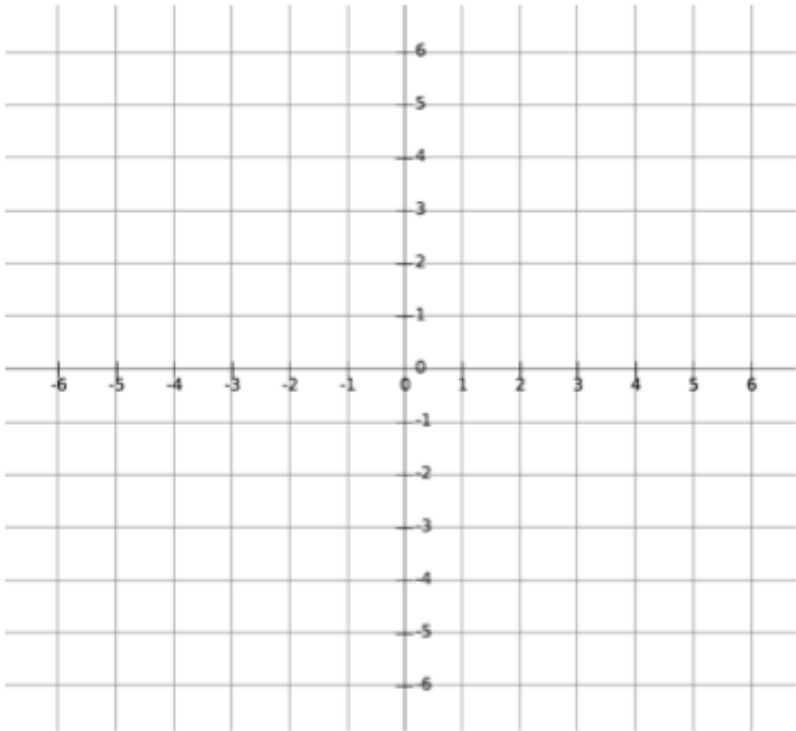
a) $g(x) = 4f(x - 3) + 0.5$

b) $g(x) = f[-2(x + 0.5)] - 1$

Example 2: for each of the following functions...

- i) make a table of values for the parent function $f(x) = \frac{1}{x}$
 - ii) describe the transformations
 - iii) make a table of values of image points
 - iv) graph the transformed function and write it's equation
- a) $g(x) = 2f(x - 1) + 2$

<i>x</i>	<i>y</i>



b) $g(x) = -f[2(x + 0.5)] - 1$

<i>x</i>	<i>y</i>

